MOS Transconductance:

As a voltage-controlled current source, a MOS transistor can be characterized by its transconductance:

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \quad \text{---- (18)}$$

This quantity serves as a measure of the "strength" of the device: a higher value corresponds to a greater change in the drain current for a given change in $V_{GS}$.

From the expression $I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$ ---(5)

we get $g_m = \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})$ ---(19)

Concluding that (1) $g_m$ is linearly proportional to $W/L$ for a given $V_{GS}-V_{TH}$, and (2) $g_m$ is linearly proportional to $V_{GS}-V_{TH}$ for a given $W/L$.

Also, substituting for $V_{GS}-V_{TH}$ from eqn (5), we obtain

$$g_m = \sqrt{2\mu_n C_{ox}\frac{W}{L} I_D} \quad \text{---- (20)}$$

This is, (1) $g_m$ is proportional to $\sqrt{W/L}$ for a given $I_D$, and (2) $g_m$ is proportional to $\sqrt{I_D}$ for a given $\frac{W}{L}$. Moreover, dividing (19) by (5)

gives

$$g_m = \frac{2I_D}{V_{GS} - V_{TH}} \quad \cdots (2)$$

revealing that (1) $g_m$ is linearly proportional to $I_D$ for a given $V_{GS} - V_{TH}$, and (2) $g_m$ is inversely proportional to $V_{GS} - V_{TH}$ for a given $I_D$. These dependencies prove critical in understanding performance trends of MOS devices and have no counterpart in bipolar transistors. Among these three expressions for $g_m$, eqn (2) is more frequently used because $I_D$ may be predetermined by power dissipation requirements.

Various dependencies of $g_m$.

| $W/L$ Constant | $W/L$ variable | $W/L$ variable |
|---|---|---|
| $V_{GS} - V_{TH}$ variable | $V_{GS} - V_{TH}$ Constant | $\frac{I_D}{V_{GS} - V_{TH}}$ Constant |
| $g_m \propto \sqrt{I_D}$ | $g_m \propto I_D$ | $g_m \propto \sqrt{\frac{W}{L}}$ |
| $g_m \propto V_{GS} - V_{TH}$ | $g_m \propto \frac{W}{L}$ | $g_m \propto \frac{1}{V_{GS} - V_{TH}}$ |

[There is some resemblance between the second column and the behavior of $g_m = I_C/V_T$. If the bipolar transistor width is increased while $V_{BE}$ remains constant, then both $I_C$ & $g_m$ increase linearly.]